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PGF5434 - Numerical Cosmology I

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1 Introduction

Cosmological structure formation describes how primordial quantum fluctuations generated during inflation evolve under gravitational instability into the complex cosmic web observed today. While linear perturbation theory provides an accurate description of density perturbations at early epochs and large spatial scales, late-time evolution on intermediate and non-linear scales requires specialized analytical approximations and numerical frameworks.

This report documents the methodology and numerical validation for the Numerical Cosmology II (EP2) project. The developed `Python` framework integrates six core cosmological pipelines:

1. **Background FLRW Dynamics & Linear Growth**: High-precision numerical ODE integration of the growth factor $D(a)$ and logarithmic growth rate $f(a)$ across flat ΛCDM, Einstein–de Sitter (EdS), and open CDM cosmologies.

2. **Gaussian Random Fields (GRF)**: Fourier-space generation of 3D linear density realizations enforcing Hermitian symmetry and exact mass conservation.

3. **Standard Perturbation Theory (SPT)**: Real-space FFT evaluation of second-order density fields $\delta_2(\mathbf{x})$, equilateral bispectrum $B(k, k, k)$, and Top-Hat smoothed reduced skewness $S_3(R)$.

4. **Lagrangian Perturbation Theory (1LPT)**: Particle displacement fields under the Zel'dovich approximation, Nearest Grid Point (NGP) mass assignment, and shot-noise-subtracted 1D power spectrum measurement.

5. **Spherical Collapse Solver**: Non-linear integration across three algebraically equivalent formulations (Models A, B, and C) and bisection shooting determination of the linear collapse threshold $\delta_c(z)$.

6. **Boltzmann Map Integration**: Transfer function reconstruction and cross-code precision benchmarking between CLASS and CAMB linear matter power spectra $P_m(k)$.

As the main reference, we follow the lecture notes from the Numerical Cosmology course (Batalha, 2026a,b,c,d,e).

2 Methodology

2.1 Cosmology and Linear Growth Dynamics

The background expansion dynamics of a general isotropic and homogeneous FLRW universe are described by the physical Hubble parameter $H(a) = H_0 E(a)$, where $H_0 \equiv 100\,h\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ is the Hubble constant and $E(a)$ is the dimensionless Hubble function. Adopting the Chevallier–Polarski–Linder (CPL) parameterization for the dark energy equation of state $w(a) = w_0 + w_a(1-a)$, $E^2(a)$ is given by

$$E^2(a) = \Omega_k(0)a^{-2} + \Omega_m(0)a^{-3} + \Omega_r(0)a^{-4} + \Omega_\Lambda(0)a^{-3(1+w_0+w_a)} \exp\left[-3w_a(1-a)\right], \quad (1)$$

where $\Omega_k(0)$, $\Omega_m(0)$, $\Omega_r(0)$, and $\Omega_\Lambda(0)$ denote the current fractional energy density parameters for spatial curvature, matter, radiation, and dark energy, respectively, satisfying $\sum \Omega_i(0) = 1$.

The scale-factor-dependent matter density parameter $\Omega_m(a) \equiv \rho_m(a)/\rho_c(a)$ is defined as

$$\Omega_m(a) = \frac{\Omega_m(0)}{E^2(a)}\, a^{-3}. \quad (2)$$

In the late-time matter-dominated era, pressureless linear perturbations evolve under the competition between gravitational instability and Hubble friction. Defining the logarithmic scale factor $N \equiv \ln a$ as the independent variable, the linear growth function $G(N)$ obeys the second-order ordinary differential equation (ODE)

$$G_{NN} + A_H(N)\, G_N - \frac{3}{2}\Omega_m(N)\, G = 0, \quad (3)$$

where $G_N \equiv \mathrm{d}G/\mathrm{d}N$, $G_{NN} \equiv \mathrm{d}^2G/\mathrm{d}N^2$, and $A_H(N)$ represents the Hubble drag parameter

$$A_H(N) \equiv 2 + \frac{\mathrm{d}\ln H}{\mathrm{d}N} = 2 + \frac{a}{2E^2(a)}\frac{\mathrm{d}E^2}{\mathrm{d}a}. \quad (4)$$

Alternatively, Eq. (3) can be cast directly in terms of the scale factor a

$$D_{aa} + \left(\frac{3}{a} + \frac{\mathrm{d}\ln H}{\mathrm{d}a}\right) D_a - \frac{3}{2}\frac{\Omega_m(a)}{a^2}\, D = 0, \quad (5)$$

where $D_a \equiv \mathrm{d}D/\mathrm{d}a$ and $D_{aa} \equiv \mathrm{d}^2D/\mathrm{d}a^2$.

2.1.1 Initial Conditions and Growth Observables

Integration starts at an early epoch deep in the matter-dominated regime, where curvature and dark energy are negligible ($\Omega_m(a_i) \approx 1$). Under pure matter domination, the growing-mode solution behaves as $G(a) \propto a$, yielding exact growing-mode initial conditions

$$G(a_i) = a_i, \qquad G_N(a_i) = a_i. \quad (6)$$

From the numerical solution $G(N)$, two scale-independent primary observables are extracted

1. **Normalized Linear Growth Factor** $D(a)$: Normalized to unity at the present epoch ($a = 1$)

$$D(a) = \frac{G(a)}{G(a=1)}. \quad (7)$$

2. **Logarithmic Growth Rate** $f(a)$: Defined as $f(a) \equiv \mathrm{d}\ln D/\mathrm{d}\ln a$, computed directly from the state variables without numerical differentiation of $D(a)$

$$f(a) = \frac{G_N(a)}{G(a)}. \tag{8}$$

2.1.2 Quadrature Validation Benchmark

To independently validate the adaptive ODE solver (`solve_ivp`), $G(a)$ is also evaluated via quadrature integration:

$$G_{\rm quad}(a) = \frac{5}{2}\Omega_m(0)\, E(a) \int_0^a \frac{\mathrm{d}\tilde{a}}{\tilde{a}^3 E^3(\tilde{a})}. \tag{9}$$

Comparing the ODE integration against the quadrature evaluation of Eq. (9) serves as a high-precision benchmark, establishing the baseline linear growth $D(a)$ and growth rate $f(a_i)$ required for setting up 1LPT initial conditions and spherical collapse shooting routines.

2.2 Gaussian Random Field Generation

Cosmological perturbation theory and particle simulations rely on Gaussian Random Fields (GRFs) to specify initial density perturbations $\delta_1(\mathbf{x})$. Given a baseline 3D linear power spectrum $P_L(k, a_{\rm ref})$, a discrete field is generated on a periodic N_g^3 spatial grid of physical box length L and volume $V = L^3$.

In Fourier space, discrete wavevectors are specified as $\mathbf{k} = (2\pi/L)(n_x, n_y, n_z)$ with $n_i \in [-N_g/2, N_g/2 - 1]$. The Fourier modes $\delta_1(\mathbf{k})$ are sampled independently from a Gaussian distribution with variance proportional to $VP_L(k)$

$$\delta_1(\mathbf{k}) = \sqrt{\frac{VP_L(k)}{2}}\,(g_1 + ig_2)\,, \tag{10}$$

where $g_1, g_2 \sim \mathcal{N}(0,1)$ are independent standard normal variables. To guarantee a real-valued density contrast $\delta_1(\mathbf{x})$ in real space via Inverse Fast Fourier Transform (IFFT), Hermitian symmetry is strictly enforced across the grid

$$\delta_1(-\mathbf{k}) = \delta_1^*(\mathbf{k}), \tag{11}$$

with the zero mode set to $\delta_1(\mathbf{k} = 0) = 0$ to conserve total cosmic mass.

This Gaussian random field generator provides the primary linear density realization $\delta_1(\mathbf{k})$ required for both Eulerian SPT (Section 2.3) and Lagrangian 1LPT (Section 2.4) implementations.

2.3 Standard Perturbation Theory (SPT) and Mode Coupling

Beyond the linear regime, non-linear mode coupling induces non-Gaussian signatures in the density field. In Eulerian Standard Perturbation Theory (SPT), the total density contrast up to second order is expanded as $\delta(\mathbf{x}) = \delta_1(\mathbf{x}) + \delta_2(\mathbf{x})$.

Given the primary linear field $\delta_1(\mathbf{k})$, the second-order field $\delta_2(\mathbf{x})$ is constructed in real space using FFT operations to evaluate advection and tidal shear interactions

$$\delta_2(\mathbf{x}) = \frac{17}{21}\delta_1^2(\mathbf{x}) + \mathbf{\Psi}^{(1)}(\mathbf{x}) \cdot \nabla\delta_1(\mathbf{x}) + \frac{2}{7}s_{ij}^2(\mathbf{x}), \tag{12}$$

where $s_{ij} = (\nabla_i \nabla_j \nabla^{-2} - 1/3\delta_{ij})\,\delta_1$ is the linear tidal shear tensor, and $\mathbf{\Psi}^{(1)} = -\nabla\nabla^{-2}\delta_1$ is the first-order displacement vector.

Non-Gaussian statistics induced by mode coupling are quantified via two primary observables

1. **Equilateral Bispectrum** $B(k,k,k)$: Measured from shell-filtered Fourier modes. At tree-level, the equilateral bispectrum target in Einstein–de Sitter (EdS) is given by
$$B^{\rm equil}_{\rm tree}(k) = \frac{12}{7} P_L^2(k). \tag{13}$$

2. **Smoothed Reduced Skewness** $S_3(R)$: Evaluated by smoothing the combined field $\delta(\mathbf{x})$ with a spherical Top-Hat window function $W_R(k) = 3(\sin kR - kR\cos kR)/(kR)^3$ of radius R
$$S_3(R) \equiv \frac{\langle \delta_R^3 \rangle}{\langle \delta_R^2 \rangle^2}. \tag{14}$$

2.4 Lagrangian Perturbation Theory (LPT)

Using the Gaussian field defined in the last section, we can construct the elements for the description of the first order of the Lagrangian perturbation, using the Zel'dovich approximation

2.4.1 Zel'dovich approximation

The Zel'dovich approximation extrapolates the first-order growing displacement, and the density is considered infinitesimal (Batalha, 2026c; White, 2014). This approximation is possible to describe from the displacement and the Lagrangian coordinate:

$$\mathbf{x}_{ZA}(\mathbf{q}, a) = \mathbf{q} - D_1(a)\nabla_{\mathbf{q}}\phi^{(1)}(\mathbf{q}) \tag{15}$$

and the velocity can be described in the same way

$$\mathbf{v}_{ZA} = -\mathcal{H} f_1 D_1 \nabla_{\mathbf{q}}\phi^{(1)} \tag{16}$$

consequently, describing the equations using the displacement, defined as

$$\mathbf{\Psi}^{(1)}_{\mathbf{k}}(a_i) = iD_1(a_i)\frac{\mathbf{k}}{k^2}\delta_{{\rm ref},\mathbf{k}} \tag{17}$$

with the following terms:

- $\mathbf{\Psi}^{(1)}_{\mathbf{k}}(a_i)$: First-order displacement field in Fourier space, evaluated at the initial scale factor a_i.
- $D_1(a_i)$: Linear growth factor evaluated at the initial scale factor
- $\mathbf{k}$: Wavevector specifying the direction aofFourier mode.
- $k = |\mathbf{k}|$: Magnitude of the wavevector.
- $\delta_{{\rm ref},\mathbf{k}}$: Fourier-space density contrast at the reference epoch.

The following expressions for the velocity and position of the particles are defined with the displacement:

$$\mathbf{x}_p = \mathbf{q}_p + \mathbf{\Psi}_p^{(1)}, \qquad \mathbf{v}_p = \mathcal{H}_i f_{1,i} \mathbf{\Psi}_p^{(1)} \tag{18}$$

From this definition, we assign particles to a mesh and compute the power spectrum:

$$P_b(k) = \frac{1}{N_k V} \sum_{\mathbf{k}_a \in S_k} |\delta_{\mathbf{k}_a}|^2 \tag{19}$$

2.4.2 Second order of LPT

In the second-order LPT, the displacement is expanded to englobe nonlinear effects that arise as cosmic structures evolve. So, the comoving position of the particle is, in this case

$$\mathbf{x}(\mathbf{q}, a) = \mathbf{q} + \mathbf{\Psi}^{(1)}(\mathbf{q}, a) + \mathbf{\Psi}^{(2)}(\mathbf{q}, a) \tag{20}$$

and in the same way for the first order, the displacement field can be expressed as

$$\mathbf{\Psi}^{(2)} = -D_2(a) \nabla_{\mathbf{q}} \phi^{(2)} \tag{21}$$

The second-order potential is determined by the first-order potential through the Poisson equation

$$\nabla^2_{\mathbf{q}} \phi^{(2)} = \frac{1}{2} [(\nabla^2_{\mathbf{q}} \phi^{(1)})^2 - \phi^{(1)}_{,ij} \phi^{(1)}_{,ij}] \tag{22}$$

with $\phi^{(1)}_{,ij} \phi^{(1)}_{,ij}$ summing the squares of the elements of the Hessian matrix, which describes how the first-order displacement varies in different directions.

The second-order growth equation is described by the following equation

$$D_2'' + \mathcal{H} D_2' - \frac{3}{2} \mathcal{H}^2 \Omega_m D_2 = -\frac{3}{2} \mathcal{H}^2 \Omega_m D_1^2, \tag{23}$$

where primes denote derivatives with respect to the conformal time τ. This equation shows the source of the second-order perturbation on the right-hand side, and the left side describes the operator governing the growth of a perturbation.

2.5 Spherical Collapse

The spherical collapse describes the evolution of an initial overdense region in the expanding background (Batalha, 2026d; Pace et al., 2017). We then follow the nonlinear evolution by solving the equations implemented in the spherical-collapse solver. We denote the initial density contrast by (δ_i) and the initial scale factor by (a_i).

In this exercise, we are using three different collapse prescriptions, as follows:

$$\text{A:} \quad \delta_{NN} + A_H \delta_N - \frac{4}{3} \frac{\delta_N^2}{1+\delta} - \frac{3}{2} \Omega_m \delta (1+\delta) = 0 \tag{24}$$

$$\text{B:} \quad \delta_N = -(1+\delta)\vartheta, \qquad \vartheta_N = -A_H \vartheta - \frac{\vartheta^2}{3} - \frac{3}{2} \Omega_m \delta \tag{25}$$

$$\text{C:} \quad g_N = g(1+g)\vartheta, \qquad \vartheta_N = -A_H \vartheta - \frac{\vartheta^2}{3} - \frac{3\Omega_m}{2g}, \tag{26}$$

with

$$A_H(N) = 2 + \frac{d \ln H}{dN}, \qquad \vartheta = \frac{\nabla_x \cdot \mathbf{v}}{aH}, \qquad g = \frac{1}{\delta}. \tag{27}$$

2.5.1 Initial conditions, turnaround and virialization

First, we describe the initial conditions at an early scale factor a_i, where the perturbation is assumed sufficiently small for linear theory to reliably describe its initial evolution. With the early factor, we compute the growth rate and the initial overdensity, determined numerically so that the nonlinear evolution reaches the prescribed collapse redshift (z_c).

An important variable is the critical linear overdensity, defined as the density contrast extrapolated to the collapse redshift. In the numerical implementation, the initial overdensity is adjusted iteratively until the nonlinear evolution satisfies the adopted collapse criterion at the target redshift. For each collapse prescription, the initial overdensity is adjusted iteratively within a prescribed search interval until the target collapse condition is satisfied.

For flat ΛCDM cosmologies, the virial radius is obtained from the energy-based virialization condition. The dimensionless virial radius is defined as $y_{\rm vir} = R_{\rm vir}/R_{\rm ta}$, where $R_{\rm ta}$ and $R_{\rm vir}$ denote the physical radii at turnaround and virialization, respectively. The implementation solves the cubic equation

$$4\lambda_{\rm ta} y_{\rm vir}^3 - 2(1+\lambda_{\rm ta}) y_{\rm vir} + 1 = 0, \tag{28}$$

where λ_{ta} is the ratio of the cosmological constant energy density to the mean density of the overdense region at turnaround. The virial overdensity is calculated using the turnaround overdensity, the ratio between the turnaround and collapse scale factors, and the dimensionless virial radius. With $a_c = (1+z_c)^{-1}$, the overdensity relative to the background matter density at collapse is

$$\Delta_{\rm vir,m}(a_c) = \zeta_{\rm ta} \left(\frac{a_c}{a_{\rm ta}}\right)^3 y_{\rm vir}^{-3}, \tag{29}$$

$$\Delta_{\rm vir,c}(a_c) = \Omega_m(a_c)\, \Delta_{\rm vir,m}(a_c). \tag{30}$$

2.6 Boltzmann Solvers and Primordial Transferring Mapping

The primordial scalar perturbation spectrum generated during inflation is parametrized by amplitude A_s and spectrum index n_s at a pivot scale $k_* = 0.05\,{\rm Mpc}^{-1}$

$$P_{\mathcal{R}}(k) = \frac{2\pi^2}{k^3} A_s \left(\frac{k}{k_*}\right)^{n_s-1}.$$

To map the primordial fluctuations to the late-time matter density perturbation $\delta_m(k,z)$, a linear Boltzmann solver integrates the coupled Einstein–Boltzmann system. Given the matter density transfer function $T_m(k,z) \equiv \delta_m(k,z)/\mathcal{R}(k)$ the linear matter power spectrum is reconstructed as

$$P_m^{\rm rec}(k,z) = P_{\mathcal{R}}(k)\, T_m^2(k,z) = \frac{2\pi^2}{k^3} A_s \left(\frac{k}{k_*}\right)^{n_s-1} T_m^2(k,z). \tag{31}$$

2.6.1 Supplementary Check A: Collisionless Boltzmann Hierarchy Closure

For a source-free, massless relativistic fluid, the multipole moments $\Psi_l(k,\tau)$ obey the infinite hierarchy of coupled differential equations

$$\dot{\Psi}_0 = -k\Psi_1, \qquad \dot{\Psi}_l = \frac{k}{2l+1}\left[l\Psi_{l-1} - (l+1)\Psi_{l+1}\right] \quad (l \geq 1). \tag{32}$$

Truncating the hierarchy at a finite multipole $l_{\max}$ introduces artificial boundary reflection unless an outgoing closure scheme is imposed

$$\Psi_{l_{\max}+1}^{(\text{outgoing})} = \frac{2l_{\max}+1}{k\tau}\Psi_{l_{\max}} - \Psi_{l_{\max}-1}. \tag{33}$$

2.6.2 Supplementary Check B: Radiation-to-Matter Superhorizon Potential Ratio

In the superhorizon ($k\mathcal{H} \ll 1$), adiabatic limit with negligible anisotropic stress, the gravitational potential $\Phi(y)$ evolves as a function of $y \equiv a/a_{\text{eq}}$ according to

$$\frac{\mathrm{d}\Phi}{\mathrm{d}\ln y} + \frac{6+5y}{2(1+y)}\Phi = -\frac{4+3y}{2(1+y)}, \tag{34}$$

with early radiation-era initial condition $\Phi_i(y_i \ll 1) = -2/3 + y_i/24$. In the late matter era ($y \gg 1$), the analytical asymptotic solution approaches $\Phi_f = -3/5$, establishing the universal superhorizon potential ratio

$$\frac{\Phi_m}{\Phi_r} = \frac{-3/5}{-2/3} = \frac{9}{10} = 0.900000. \tag{35}$$

3 Numerical Checks

3.1 Cosmology Validation (ex 1)

To validate the numerical implementation of the background FLRW dynamics and linear growth solver, we tested three distinct cosmological models

1. **Einstein–de Sitter (EdS):** Pure matter-dominated universe ($\Omega_m(0) = 1.0$, $\Omega_\Lambda(0) = 0.0$).
2. **Flat ΛCDM (Fiducial):** $\Omega_m(0) = 0.315$ ($\Omega_b(0) = 0.0493$, $\Omega_{\rm cdm}(0) = 0.2657$), $\Omega_\Lambda(0) = 0.685$, $h = 0.6737$, $w_0 = -1.0$, $w_a = 0.0$.
3. **Open CDM (oCDM):** $\Omega_m(0) = 0.315$, $\Omega_\Lambda(0) = 0.0$, $\Omega_k(0) = 0.685$.

3.1.1 Linear Growth Observables across Cosmological Models

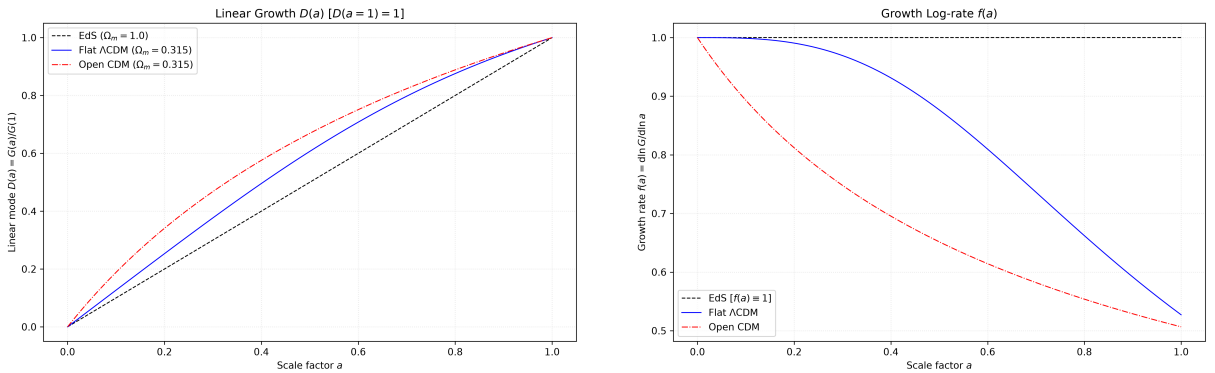
Table 1 summarizes the normalized linear growth factor $D(a = 1)$ and logarithmic growth rate $f(a = 1) \equiv \mathrm{d}\ln D/\mathrm{d}\ln a$ obtained at the present epoch $a = 1$.

Table 1: Linear growth observables $D(a = 1)$ and $f(a = 1)$ computed across cosmological models integrated from $a_i = 10^{-4}$. EdS recovers the exact analytical limit $D = 1$ and $f = 1$.

Model	$\Omega_{m,0}$	$\Omega_{\Lambda,0}$	$D(a = 1)$	$f(a = 1)$
EdS (Analytical Benchmark)	1.000	0.000	1.000000	1.000000
Flat ΛCDM (Fiducial)	0.315	0.685	1.000000	0.527103
Open CDM (oCDM)	0.315	0.000	1.000000	0.506548

The numerical integration reproduces the exact theoretical limit for Einstein–de Sitter ($D(a) = a$, $f(a) = 1$) with zero error at $a = 1$. In ΛCDM and oCDM, dark energy acceleration and spatial curvature suppress structure growth at late times ($a \gtrsim 0.5$), reducing the growth rate today to $f(1) \approx 0.5271$ and $f(1) \approx 0.5065$, respectively.

Figure 1 presents the scale factor evolution of $D(a)$ and $f(a)$ for all three cosmologies.



(a) Normalized linear growth factor $D(a)$.

(b) Logarithmic growth rate $f(a)$.

Figure 1: Evolution of linear growth observables $D(a)$ and $f(a)$ comparing EdS, flat ΛCDM, and open CDM models from $a_i = 10^{-4}$ to $a = 1.0$.

3.1.2 Quadrature vs. ODE Solver Benchmark

To independently confirm the accuracy of the adaptive ODE solver (`solve_ivp`), the unnormalized growth function $G(a)$ was evaluated via quadrature integration. The relative error between the ODE solution $G_{\rm ODE}(a)$ and the quadrature integration $G_{\rm quad}(a)$ across the interval $a \in [10^{-4}, 1.0]$ was measured at

$$\frac{|G_{\rm ODE}(a=1) - G_{\rm quad}(a=1)|}{G_{\rm quad}(a=1)} = 2.124 \times 10^{-7}, \tag{36}$$

confirming excellent agreement between the two independent integration methods, as shown in Figure 2.

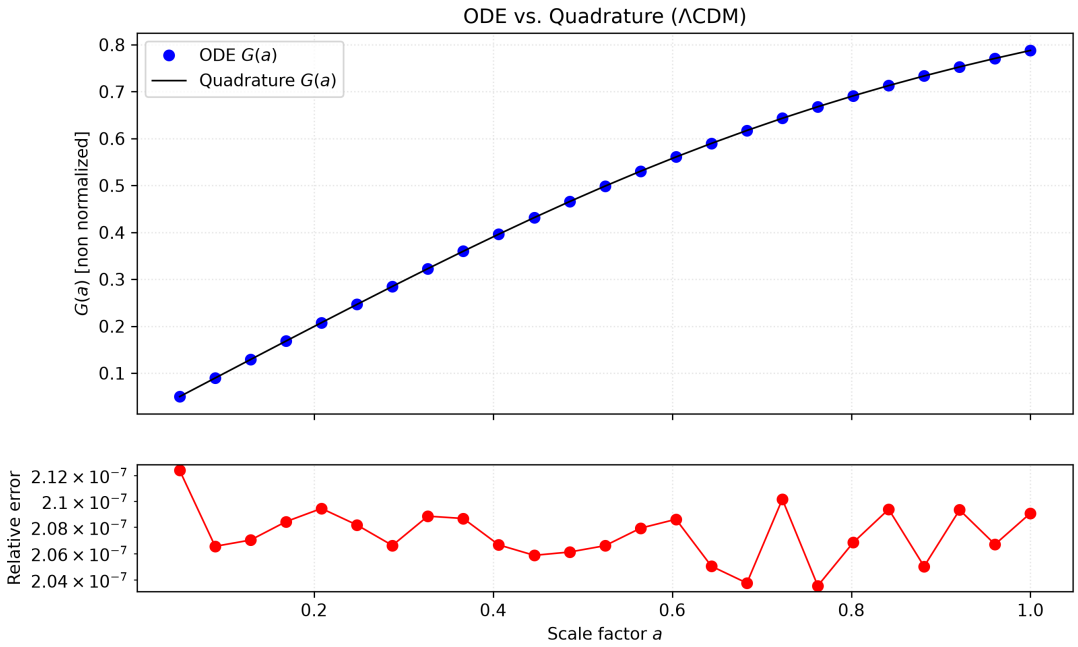


Figure 2: Relative error profile comparing $G(a)$ computed via ODE integration (`solve_ivp`) against quadrature evaluation across $a \in [10^{-4}, 1.0]$.

3.1.3 Initial Scale Factor and Solver Tolerance Sensitivity Analysis

To separate initial-mode transient errors from ODE solver tolerances, we systematically varied the starting scale factor $a_i \in \{10^{-3}, 10^{-4}, 10^{-5}\}$ and adaptive integration tolerances $(\text{rtol}, \text{atol}) \in \{(10^{-6}, 10^{-8}), (10^{-8}, 10^{-10}), (10^{-10}, 10^{-12})\}$.

Table 2 displays the stability of the observables $D(a = 1)$ and $f(a = 1)$ in flat ΛCDM under these varying numerical controls.

The sensitivity study demonstrates that the normalized growth factor $D(a = 1) = 1.00000000$ remains invariant across all test conditions; the logarithmic growth rate converges to $f(a = 1) = 0.52710320$ once tolerances reach rtol $\leq 10^{-8}$, with relative variations bounded below 10^{-6} across three orders of magnitude in a_i; and starting at $a_i \leq 10^{-4}$ with tolerances $(\text{rtol}, \text{atol}) = (10^{-8}, 10^{-10})$ eliminates initial-mode transient contamination, providing a stable baseline for subsequent 1LPT and spherical collapse computations.

Table 2: Sensitivity analysis of growth observables $f(a=1)$ and $D(a=1)$ in flat ΛCDM as a function of initial scale factor a_i and integration tolerances (rtol, atol). Relative errors in $f(1)$ are computed against the highest precision baseline.

a_i	**Tolerances** (rtol, atol)	$f(a=1)$	$D(a=1)$	**Rel. Error in** $f(1)$
1.0×10^{-3}	$(10^{-6}, 10^{-8})$	0.52710275	1.00000000	0.00×10^{0}
1.0×10^{-3}	$(10^{-8}, 10^{-10})$	0.52710320	1.00000000	8.54×10^{-7}
1.0×10^{-3}	$(10^{-10}, 10^{-12})$	0.52710320	1.00000000	8.62×10^{-7}
1.0×10^{-4}	$(10^{-6}, 10^{-8})$	0.52710281	1.00000000	1.24×10^{-7}
1.0×10^{-4}	$(10^{-8}, 10^{-10})$	0.52710320	1.00000000	8.54×10^{-7}
1.0×10^{-4}	$(10^{-10}, 10^{-12})$	0.52710320	1.00000000	8.62×10^{-7}
1.0×10^{-5}	$(10^{-6}, 10^{-8})$	0.52710276	1.00000000	2.71×10^{-8}
1.0×10^{-5}	$(10^{-8}, 10^{-10})$	0.52710320	1.00000000	8.54×10^{-7}
1.0×10^{-5}	$(10^{-10}, 10^{-12})$	0.52710320	1.00000000	8.62×10^{-7}

3.2 Standard Perturbation Theory Validation (ex 2)

To validate the second-order Standard Perturbation Theory (SPT) module, numerical tests were conducted on an $N_g^3 = 128^3$ grid with physical box length $L = 500\,h^{-1}$Mpc under two distinct linear variance regimes:

1. **Strong Coupling Regime ($\sigma_1^2 \approx 0.4613$):** Designed for visualizing the spatial non-linear clustering in 2D density slices and the 1D power spectrum enhancement $P_{\rm tot}(k) = P_1(k) + P_{22}(k) + 2P_{13}(k)$, yielding a total non-linear variance of $\sigma^2_{\rm tot} \approx 0.9233$.

2. **Weakly Non-Linear Regime ($\sigma_1^2 \approx 4.6116 \times 10^{-3}$):** Adopted for rigorous perturbative comparison of the bispectrum and reduced skewness against tree-level analytical predictions, avoiding higher-order breakdown.

3.2.1 Density Slices and 1D Power Spectrum Enhancement

Figure 3 shows the spatial density contrast, and Figure 4 shows the 1D power spectrum generated in the strong coupling regime.

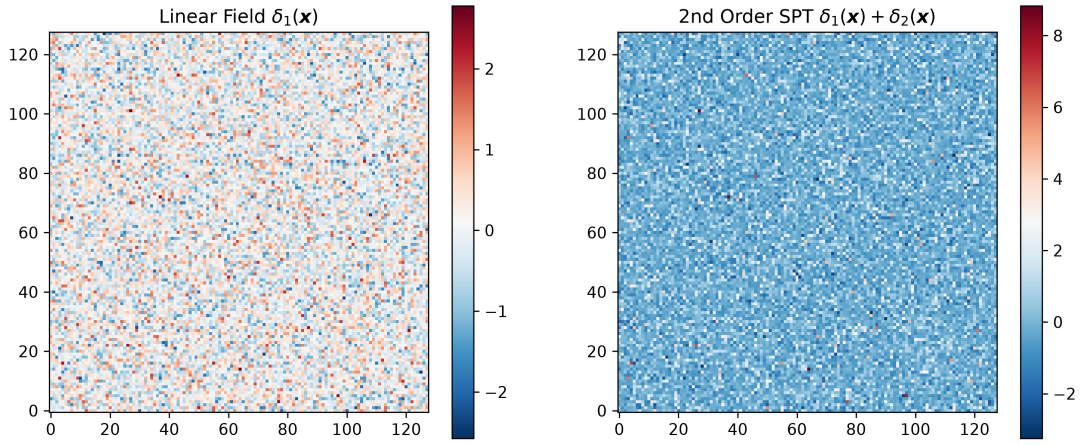


Figure 3: Linear density $\delta_1(\mathbf{x})$ vs. 2nd-order SPT $\delta_1(\mathbf{x}) + \delta_2(\mathbf{x})$.

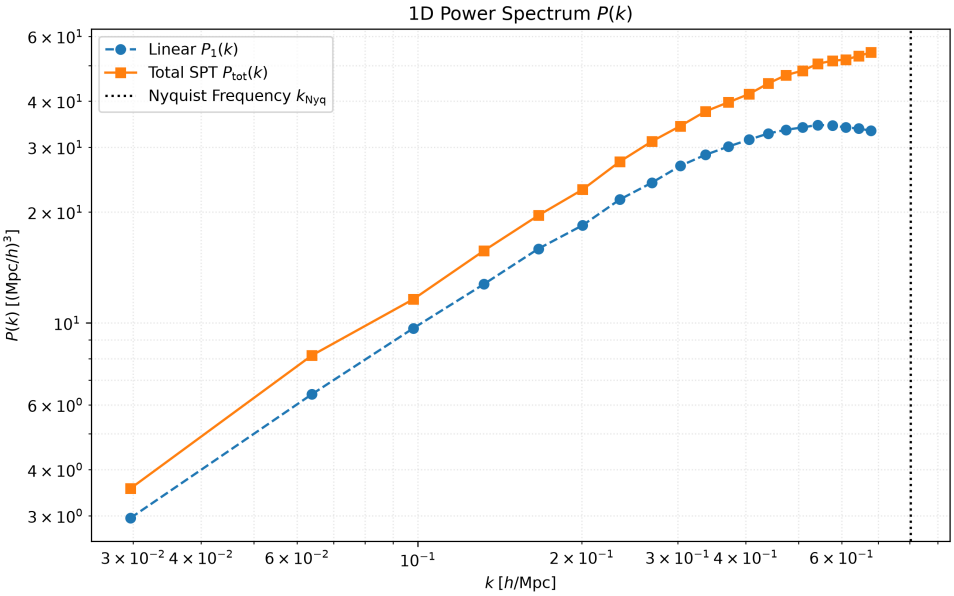


Figure 4: 1D matter power spectrum $P_1(k)$ vs. $P_{\rm tot}(k)$.

Advection and tidal shear interactions in $\delta_2(\mathbf{x})$ assemble mass into localized, high-density structures (Figure 3), resulting in systematic power enhancement in $P_{\rm tot}(k)$ relative to linear theory across all resolved scales up to $k_{\rm Nyq}$ (Figure 4).

3.2.2 Higher-Order Non-Gaussian Statistics

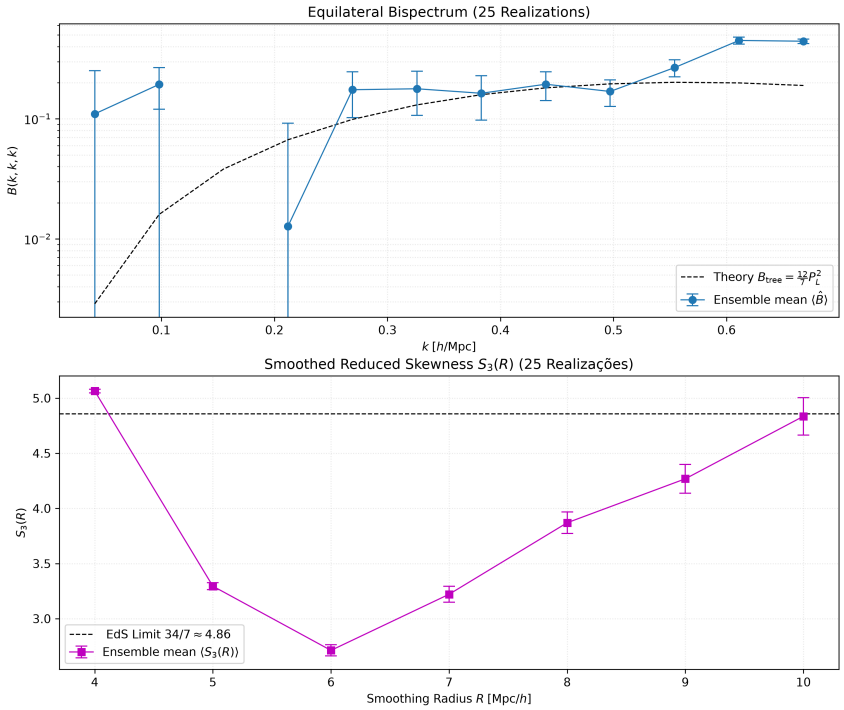


Figure 5: Equilateral bispectrum $\langle\hat{B}(k,k,k)\rangle$ and smoothed reduced skewness $\langle S_3(R)\rangle$ averaged over 25 independent realizations ($\sigma_1^2 = 4.6116\times10^{-3}$), demonstrating convergence to tree-level theory $B_{\rm tree} = 12P_L^2/7$ and the Einstein–de Sitter limit $S_3 = 34/7 \approx 4.86$.

In the weakly non-linear regime ($\sigma_1^2 \approx 4.6116 \times 10^{-3}$), the equilateral bispectrum $B(k, k, k)$ across 12 wave-number bins and the Top-Hat smoothed reduced skewness $S_3(R)$ across $R \in [4, 12]\, h^{-1}$Mpc were evaluated.

As shown in Figure 5, averaging the equilateral bispectrum over an ensemble of $N_{\rm real} = 25$ independent realizations reduces single-box sampling scatter, matching the tree-level SPT expectation $B_{\rm tree}^{\rm equil}(k) = (12/7)P_L^2(k)$ across intermediate modes ($0.25\, h/\text{Mpc} \leq k \leq 0.50\, h/\text{Mpc}$). Furthermore, the reduced skewness profile demonstrates that the ensemble mean $\langle S_3(R) \rangle$ recovers the exact analytical Einstein–de Sitter value $S_3^{\rm EdS} = 34/7 \approx 4.86$ at smoothing radii $R = 4\, h^{-1}$Mpc and $R = 10\, h^{-1}$Mpc, validating the non-Gaussian mode coupling implementation.

3.3 Lagrangian Perturbation Theory Validation (ex 3)

3.3.1 First-Order LPT

The first step in validating the numerical implementation of First-Order Lagrangian Perturbation Theory (1LPT) is to examine the structure formation predicted by the Zel'dovich approximation (ZA). This provides a qualitative check of the particle displacement field and its ability to reproduce the large-scale structures generated by gravitational evolution.

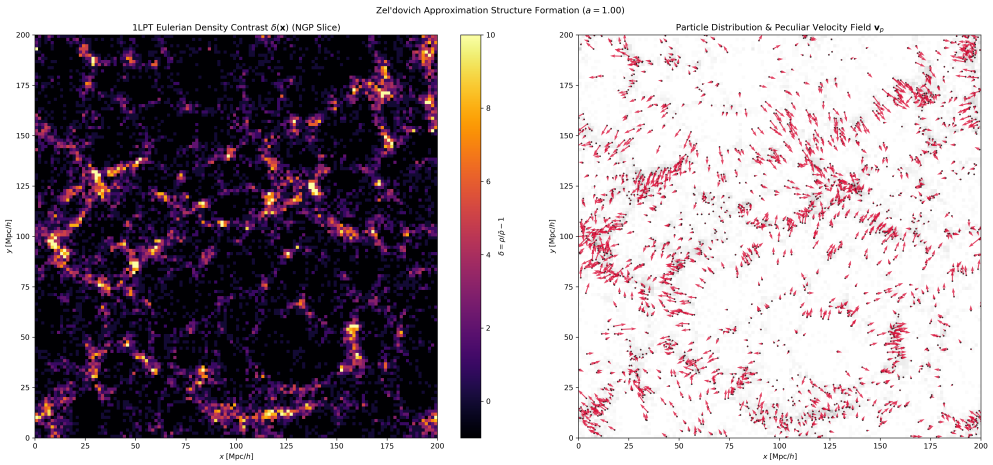


Figure 6: Cosmic web structure produced by the numerical implementation of the Zel'dovich approximation.

Figure (6) shows the formation of filamentary and clustered structures characteristic of the cosmic web. This qualitative agreement provides an initial indication that the displacement field behaves as expected. Therefore, we also compare the measured matter power spectrum with the linear power spectrum generated by CAMB, using the reference linear spectrum $P_{\rm L}(k, a_{\rm ref})$ and linear growth factor $D_1(a_{\rm ref})$.

The first comparison examines the measured power spectra with the linear power spectrum:

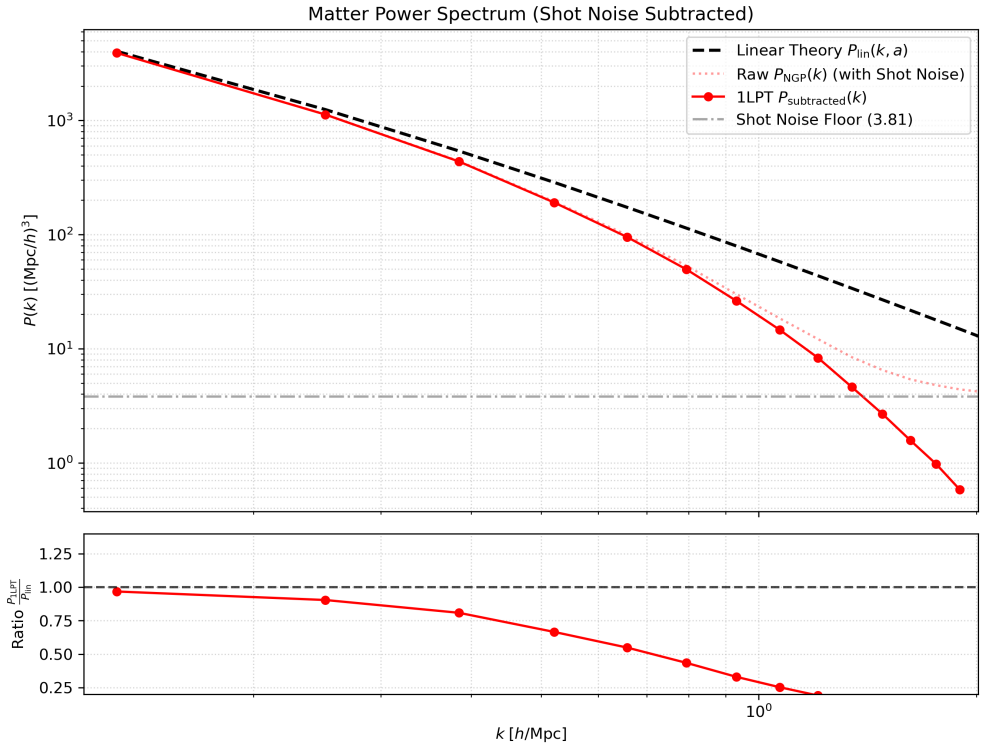


Figure 7: Comparison between the linear matter power spectrum and the power spectra measured

For a more detail analysis, we habe to compare different realizations. This allows us to investigate the dependence of the measured spectrum on the initial random realization and numerical resolution. Differences between the measured and theoretical spectra can arise from finite-volume effects, sampling variance, mass assignment onto the grid, and limitations associated with estimating power at large wavenumbers.

To obtain a more robust comparison, we average the measured spectra over ten random seeds for each grid resolution, considering $N_g = 64$, 128, and 256. We also account for the mass-assignment window function associated with depositing particles onto the grid. The squared Fourier-space window function is defined as

$$|W(\mathbf{k})|^2 = \prod_{i=x,y,z} \left[\mathrm{sinc}\left(\frac{k_i \Delta x}{2\pi} \right) \right]^{2p}, \tag{37}$$

where $\Delta x = L/N_g$ is the grid spacing, p is the order of the mass-assignment scheme, and $\mathrm{sinc}(x) = \sin(\pi x)/(\pi x)$ is the normalized sinc function. This expression describes the suppression of Fourier modes introduced by mass assignment. Since the power spectrum is proportional to the squared modulus of the Fourier-space density contrast, $P(k) \propto |\delta(\mathbf{k})|^2$, the corresponding power-spectrum window is $|W(\mathbf{k})|^2$.

When appropriate, the measured spectrum can be corrected for this effect by dividing by the squared window function. Such a correction is particularly relevant at high wavenumbers, where mass assignment can significantly suppress the measured power. However, this correction does not eliminate other sources of numerical error, such as aliasing. The resulting comparison, combining multiple realizations and grid resolutions, is shown in Figure(8):

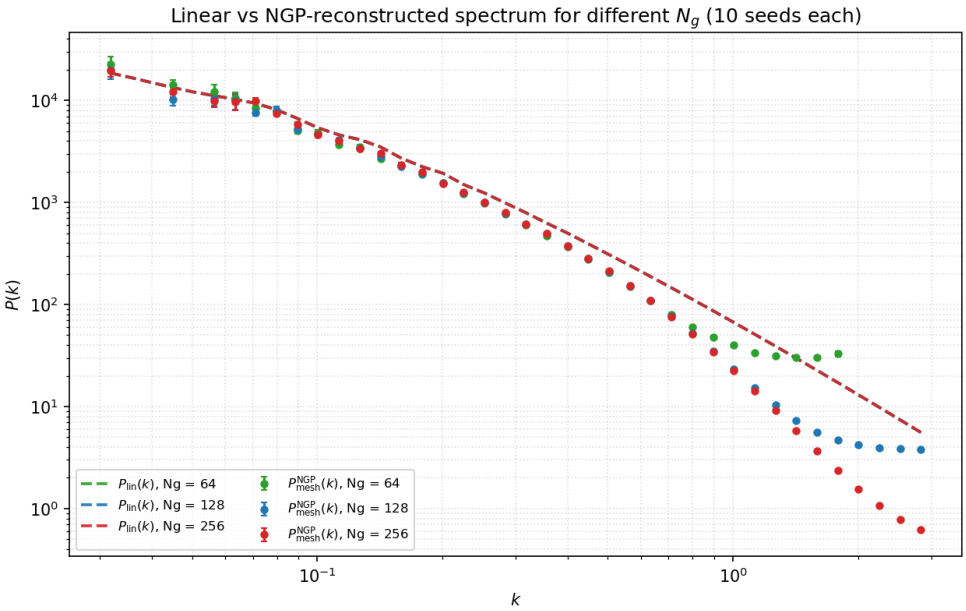


Figure 8: Comparison between the linear matter power spectrum and the averaged power spectra measured from the numerical realizations for $N_g = 64$, 128, and 256, using ten random seeds for each resolution. The mass-assignment window correction is included according to the adopted estimator.

Averaging over random seeds reduces the impact of realization-dependent fluctuations and makes the resolution dependence easier to identify. The comparison with the CAMB spectrum therefore provides a quantitative test of the initial conditions and the numerical power-spectrum estimator. In particular, agreement over the range of well-resolved wavenumbers supports the consistency of the implemented first-order displacement field and the associated density-field measurement.

3.3.2 Second-Order LPT

The numerical validation of Second-Order Lagrangian Perturbation Theory (2LPT) is performed using the bispectrum and reduced skewness estimators introduced in Exercise 2. These statistics complement the power spectrum by characterizing the non-Gaussian features generated by nonlinear gravitational evolution.

First, we measure the bispectrum for equilateral Fourier-space triangles in the initial Gaussian field, the Zel'dovich approximation, and the 2LPT output:

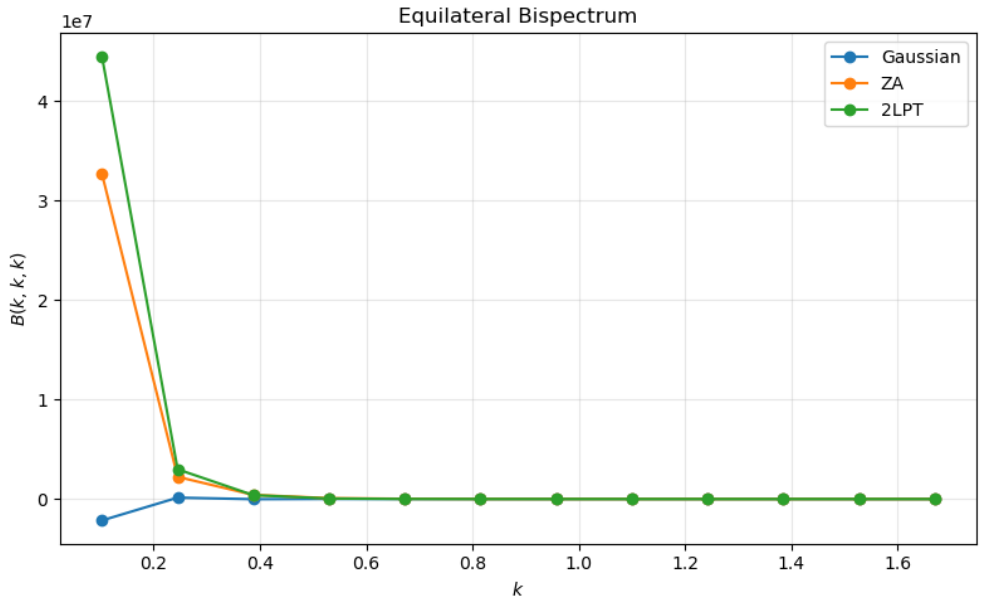


Figure 9: Comparison of the equilateral bispectrum measured from the initial Gaussian field, the Zel'dovich approximation, and the 2LPT output.

As shown in Figure (9), the initial Gaussian field exhibits a bispectrum consistent with zero within the statistical and numerical uncertainties, as expected for an ideal Gaussian random field. In contrast, the ZA and 2LPT outputs develop nonzero bispectra, reflecting the non-Gaussian density fluctuations generated by nonlinear evolution. The bispectrum therefore provides a useful diagnostic of the nonlinear mode coupling introduced by the Lagrangian displacement field.

We further investigate the non-Gaussianity of the evolved density fields by measuring the reduced skewness as a function of the smoothing scale R.

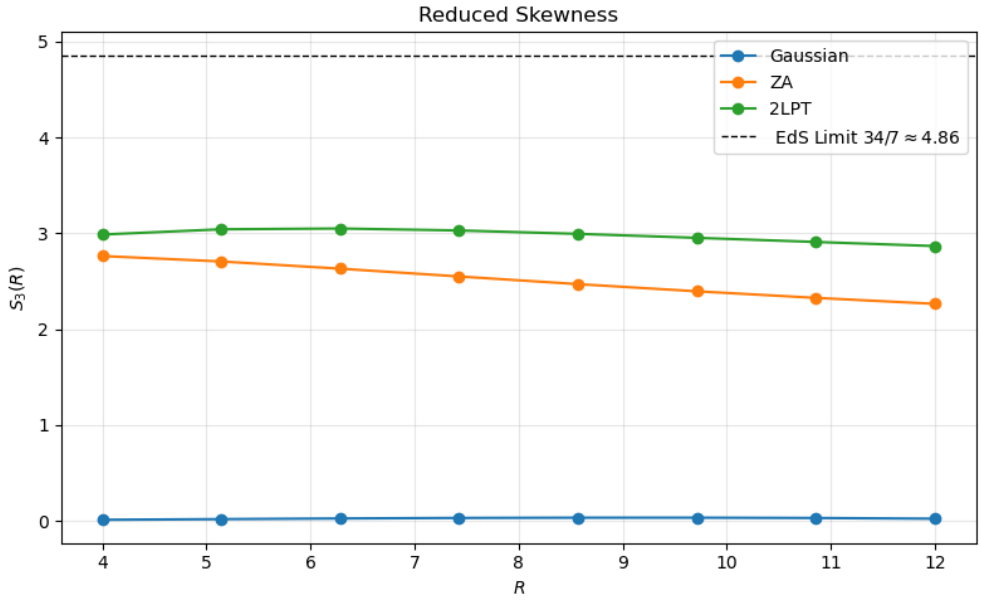


Figure 10: Comparison of the reduced skewness as a function of smoothing scale R for the initial Gaussian field, the Zel'dovich approximation, and the 2LPT output.

Figure (10) shows that the initial Gaussian field has reduced skewness close to zero, whereas the ZA and 2LPT density fields develop positive skewness. This behavior is consistent with the generation of non-Gaussian density statistics through gravitational evolution, which produces an increasingly asymmetric density distribution as overdense regions grow.

Taken together, the bispectrum and reduced-skewness measurements provide complementary tests of the 2LPT implementation. The bispectrum probes correlations between Fourier modes, while the skewness characterizes the asymmetry of the smoothed density distribution. Their comparison across the Gaussian, ZA, and 2LPT fields helps assess whether the numerical evolution produces the expected nonlinear statistical signatures.

3.4 Spherical Collapse (ex 4)

To validate the numerical implementation of the spherical collapse solver, two primary tests were performed: (i) verifying the equivalence among formulations A, B, and C under non-linear integration, and (ii) determining the linear collapse threshold $\delta_c(z)$ using a bisection shooting method across different cosmological models.

3.4.1 Equivalence of Formulations A, B, C

To assess the numerical equivalence among the three spherical collapse formulations (Models A, B, and C), each model was integrated from an initial scale factor $a_i = 10^{-4}$ ($\delta_i = 10^{-4}$) to $a_f = 1.0$ within a fiducial flat ΛCDM background ($\Omega_{m,0} = 0.315$). Table 3 summarizes the final density contrast $\delta(a_f)$ and the maximum relative differences evaluated with respect to Model A. Models B and C reproduce the reference Model A trajectory with maximum relative differences bounded at 5.52×10^{-9} and 5.79×10^{-7}, respectively, demonstrating high numerical agreement across the formulations.

Table 3: Numerical comparison and relative errors among Spherical Collapse Models A, B, and C integrated from $a_i = 10^{-4}$ ($\delta_i = 10^{-4}$) to $a_f = 1.0$ in flat ΛCDM ($\Omega_{m,0} = 0.315$).

Model	$\delta(a_i)$	$\delta(a_f)$	**Max. Rel. Diff. vs. Model A**
Model A (δ, δ_N)	1.000000×10^{-4}	1.939895	Reference
Model B (δ, θ)	1.000000×10^{-4}	1.939895	5.5159×10^{-9}
Model C (g, θ)	1.000000×10^{-4}	1.939894	5.7892×10^{-7}

Figure 11 displays the scale factor evolution of the overdensity $\delta(a)$ and the velocity divergence $\theta(a) \equiv \nabla \cdot \mathbf{v}/(aH)$. The trajectories across Models A, B, and C exhibit complete numerical overlap throughout the entire non-linear evolution up to $a = 1.0$.

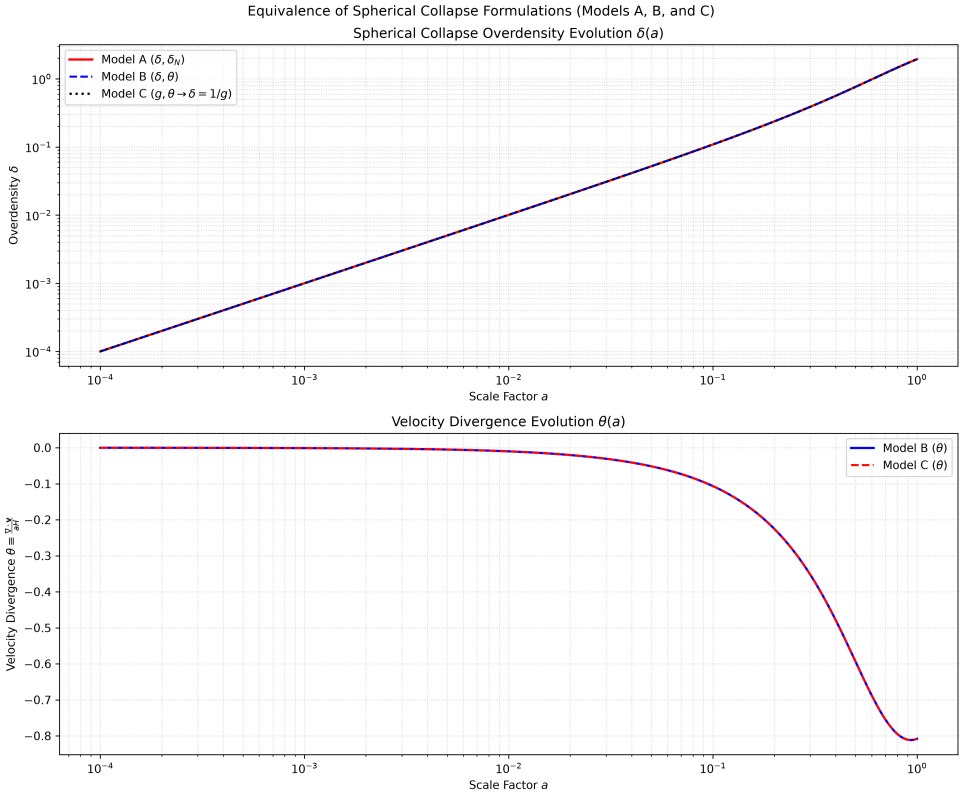


Figure 11: Evolution of the overdensity $\delta(a)$ (top) and velocity divergence $\theta(a)$ (bottom) for Models A, B, and C integrated from $a_i = 10^{-4}$ to $a_f = 1.0$, demonstrating complete numerical overlap.

3.4.2 Linear Collapse Threshold $\delta_c(z)$ and Shooting Method Validation

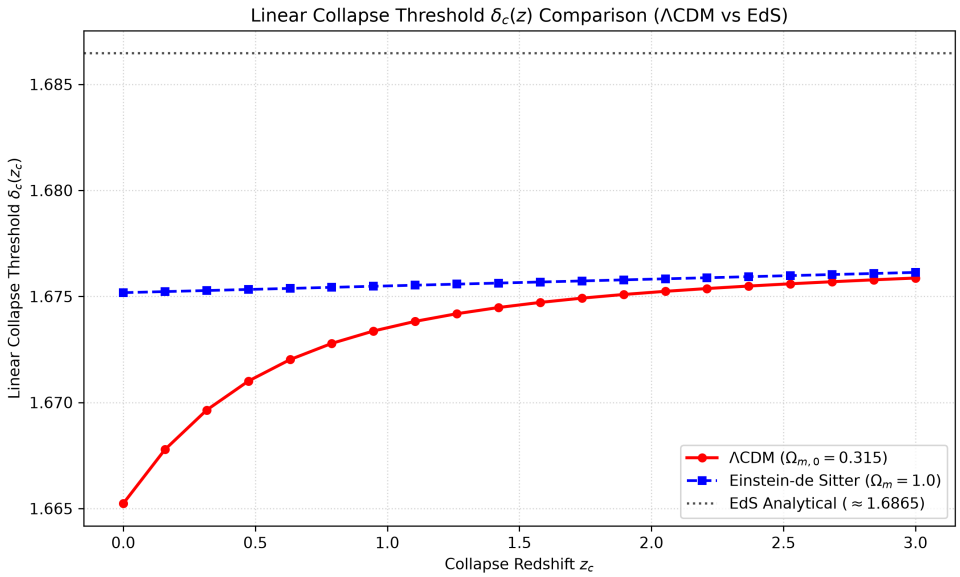


Figure 12: Linear collapse threshold $\delta_c(z_c)$ as a function of collapse redshift z_c comparing flat ΛCDM ($\Omega_{m,0} = 0.315$) and Einstein–de Sitter ($\Omega_m = 1.0$).

The critical linear collapse threshold $\delta_c(z_c)$ was determined using a bisection shooting algorithm designed to tune the initial amplitude δ_i at a_i such that the non-linear perturbation reaches formal collapse at target redshift z_c. Figure 12 illustrates the redshift evolution of $\delta_c(z_c)$ in flat ΛCDM ($\Omega_{m,0} = 0.315$) compared against the Einstein–de Sitter ($\Omega_m = 1.0$) benchmark and its analytical limit ($\delta_c \approx 1.6865$). While the EdS threshold remains nearly constant near $1.675 \to 1.6765$, dark energy domination at late times ($z_c \to 0$) in ΛCDM suppresses linear growth relative to the non-linear evolution, lowering the threshold to $\delta_c(0) = 1.665235$.

To verify that the shooting algorithm delivers formulation-independent results, $\delta_c(z_c)$ was evaluated across discrete collapse redshifts $z_c \in \{0.0, 0.5, 1.0, 2.0, 3.0\}$ for Models A, B, and C. As detailed in Table 4, the critical threshold values calculated across all three models match up to six decimal places, with relative percentage differences relative to Model A remaining below $2.8 \times 10^{-8}\%$ for Model B and $9.1 \times 10^{-8}\%$ for Model C.

Table 4: Critical linear collapse threshold $\delta_c(z_c)$ for Models A, B, and C (panel a) and the corresponding relative percentage differences with respect to Model A (panel b).

(a) Critical linear collapse threshold $\delta_c(z_c)$

z_c	**Model A (δ_c)**	**Model B (δ_c)**	**Model C (δ_c)**
0.0	1.665235	1.665235	1.665235
0.5	1.671197	1.671197	1.671197
1.0	1.673532	1.673532	1.673532
2.0	1.675192	1.675192	1.675192
3.0	1.675866	1.675866	1.675866

(b)Relative percentage difference compared to Model A (%)

z_c	**Model A (%)**	**Model B vs. A (%)**	**Model C vs. A (%)**
0.0	0.0	1.9505×10^{-8}	7.8744×10^{-8}
0.5	0.0	2.6711×10^{-8}	9.1182×10^{-8}
1.0	0.0	2.2635×10^{-8}	8.9818×10^{-8}
2.0	0.0	2.4564×10^{-8}	8.9527×10^{-8}
3.0	0.0	2.7827×10^{-8}	8.8051×10^{-8}

This cross-model consistency is illustrated in Figure 13, where the threshold curves for Models A, B, and C overlap across the entire evaluated redshift range.

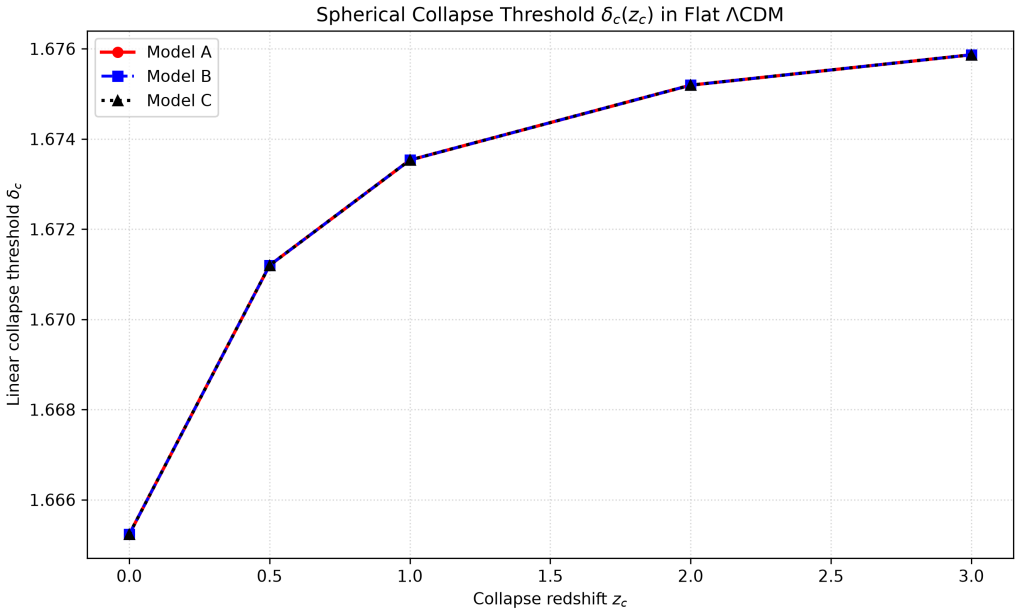


Figure 13: Linear collapse threshold $\delta_c(z_c)$ calculated across discrete redshifts $z_c \in \{0, 0.5, 1, 2, 3\}$ for Models A, B, and C, confirming cross-model consistency.

3.4.3 Cosmological Grid Analysis and Independent Audit

To systematically map spherical collapse observables, a cosmological grid comprising 63 parameter sets (3 models $\times$ 7 redshifts $z_c \in \{0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0\} \times 3$ matter densities $\Omega_{m,0} \in \{0.2, 0.315, 0.4\}$) was computed.

An independent mathematical audit confirmed the precision of all turnaround and virial observables

- The turnaround event condition $\theta_{\rm ta} + 3 = 0$ was located with residuals bounded below 3.9968×10^{-15}.
- The matter-referenced virial overdensity $\Delta_{\rm vir,m}$ formula error was bounded below 5.6843×10^{-14}.
- The critical-referenced virial overdensity $\Delta_{\rm vir,c}$ formula error was bounded below 8.5265×10^{-14}.
- The cubic virial polynomial residual $4\lambda_{\rm ta} y_{\rm vir}^3 - 2(1+\lambda_{\rm ta}) y_{\rm vir} + 1 = 0$ yielded maximum residuals of 5.0182×10^{-14}.

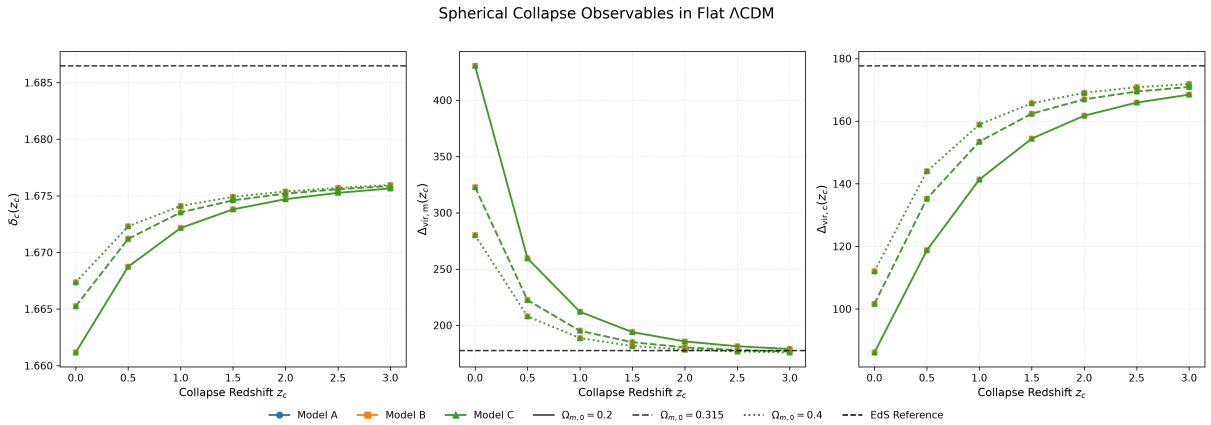


Figure 14: Spherical collapse observables $\delta_c(z_c)$ (left), $\Delta_{\mathrm{vir,m}}(z_c)$ (middle), and $\Delta_{\mathrm{vir,c}}(z_c)$ (right) computed across the 63-case cosmological grid for Models A, B, and C against Einstein–de Sitter reference values.

The diagnostic verification checklist confirmed that all 63 cases were populated with finite values. The single check flagged as `False` during initial script execution was identified as a column key string mismatch (`y_root_poly` vs. `y_root_polynomial`); the physical root $y_{\mathrm{root_poly}}$ matched y_{vir} identically across all 63 grid points.

3.4.4 Convergence and Sensitivity Analysis

Numerical convergence tests were performed in Einstein–de Sitter (EdS) and flat ΛCDM backgrounds by systematically varying

1. Initial scale factor $a_i \in \{10^{-3}, 10^{-4}, 10^{-5}\}$.
2. Singular cutoff threshold $\delta_{\mathrm{stop}} \in \{10^5, 10^6, 10^7, 10^8\}$.
3. Integration solver tolerances $(\mathrm{rtol}, \mathrm{atol}) \in \{(10^{-6}, 10^{-8}), (10^{-8}, 10^{-10}), (10^{-10}, 10^{-12})\}$.

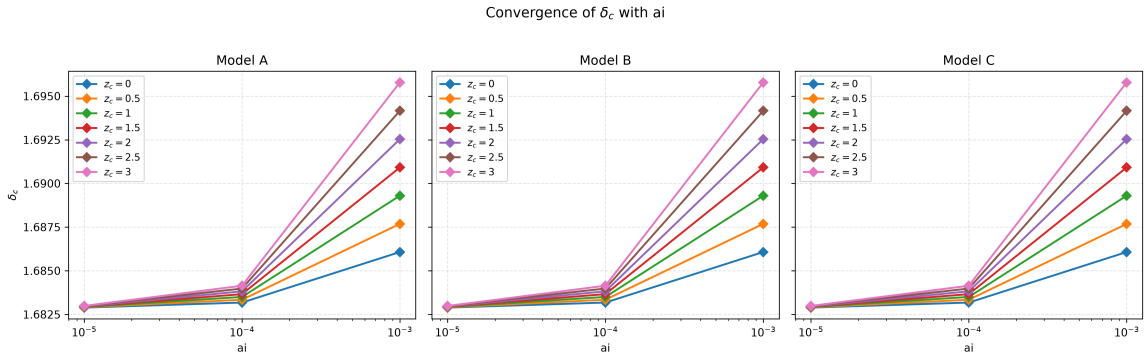


Figure 15: Convergence of the linear collapse threshold δ_c with initial scale factor a_i across Models A, B, and C.

The convergence study confirms that

- Transients from initial conditions decay rapidly for starting scale factors $a_i \leq 10^{-4}$, where δ_c and virial overdensities stabilize.

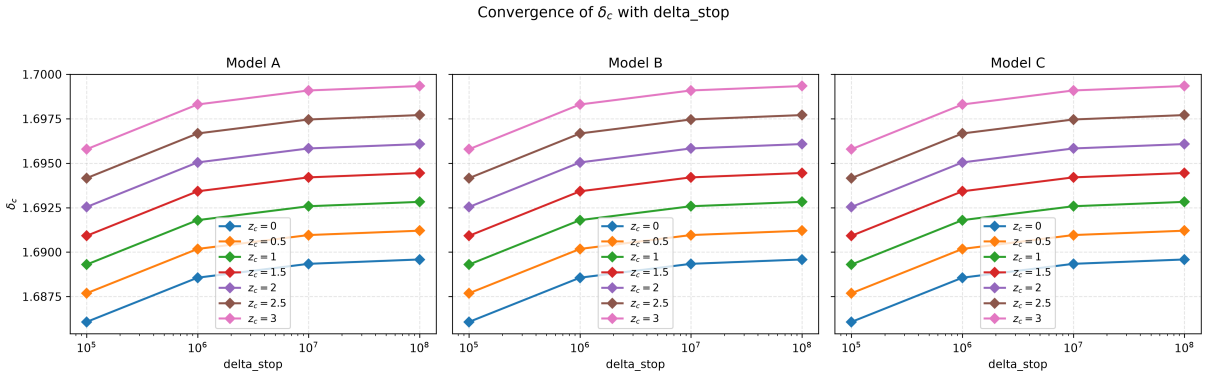


Figure 16: Convergence of δ_c with singular threshold cutoff $\delta_{\rm stop}$.

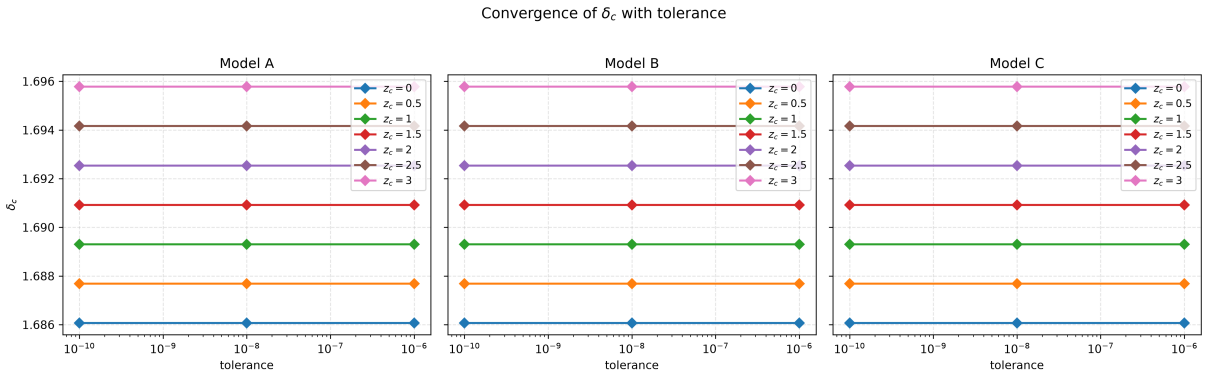


Figure 17: Dependence of δ_c on ODE integrator numerical tolerances.

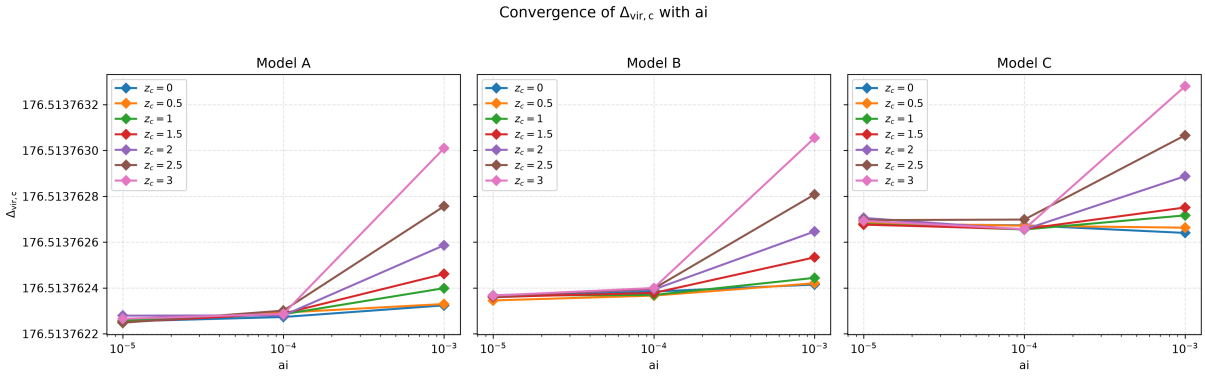


Figure 18: Convergence of the critical virial overdensity $\Delta_{\rm vir,c}$ as a function of a_i.

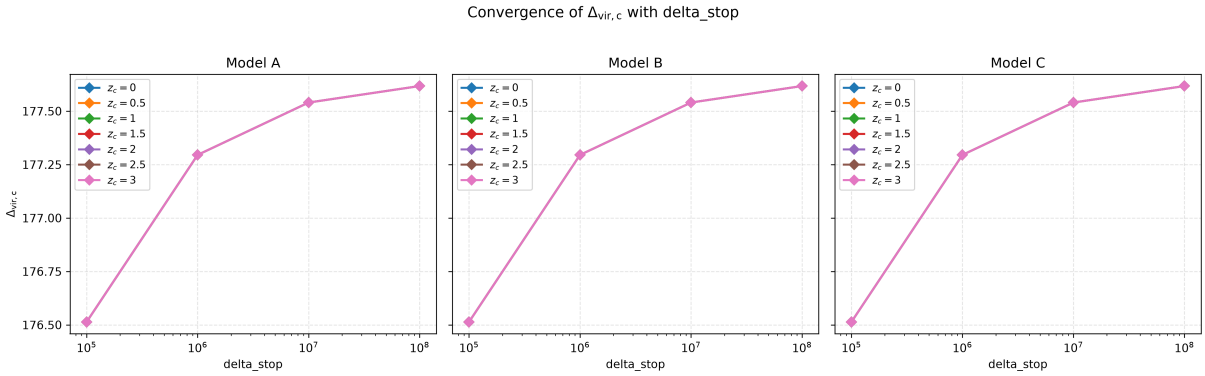


Figure 19: Convergence of $\Delta_{\rm vir,c}$ with singular cutoff $\delta_{\rm stop}$.

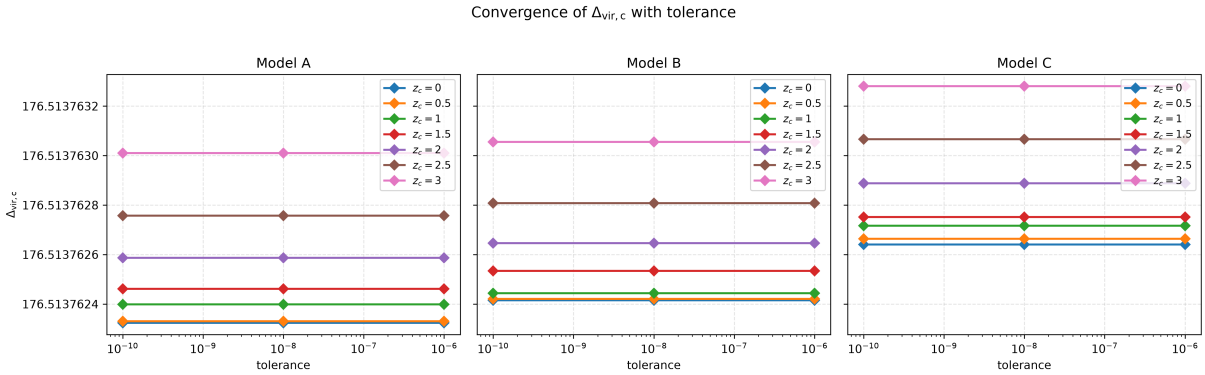


Figure 20: Dependence of $\Delta_{\mathrm{vir,c}}$ on solver integration tolerances.

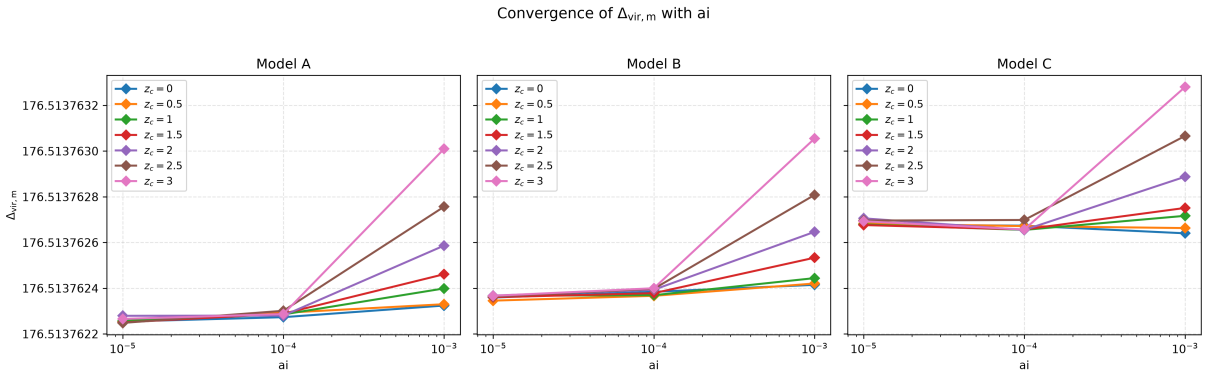


Figure 21: Convergence of the matter virial overdensity $\Delta_{\mathrm{vir,m}}$ as a function of a_i.

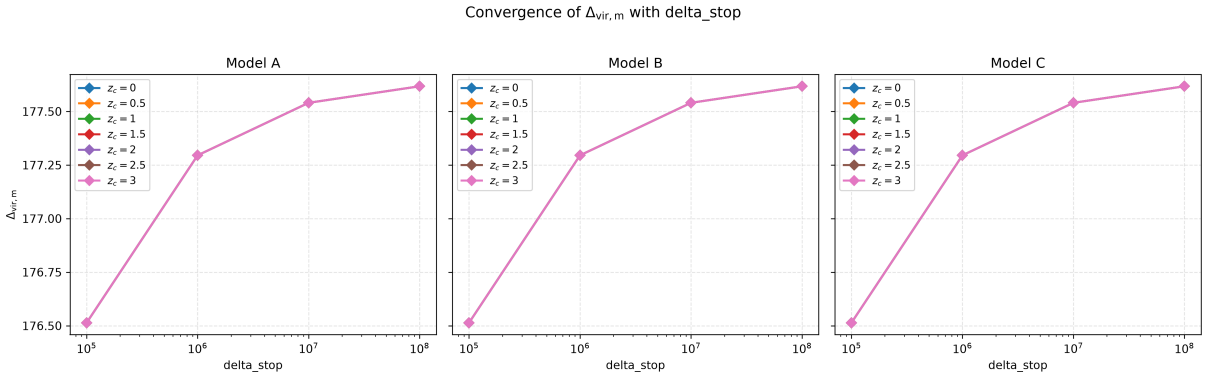


Figure 22: Convergence of $\Delta_{\mathrm{vir,m}}$ with singular cutoff δ_{stop}.

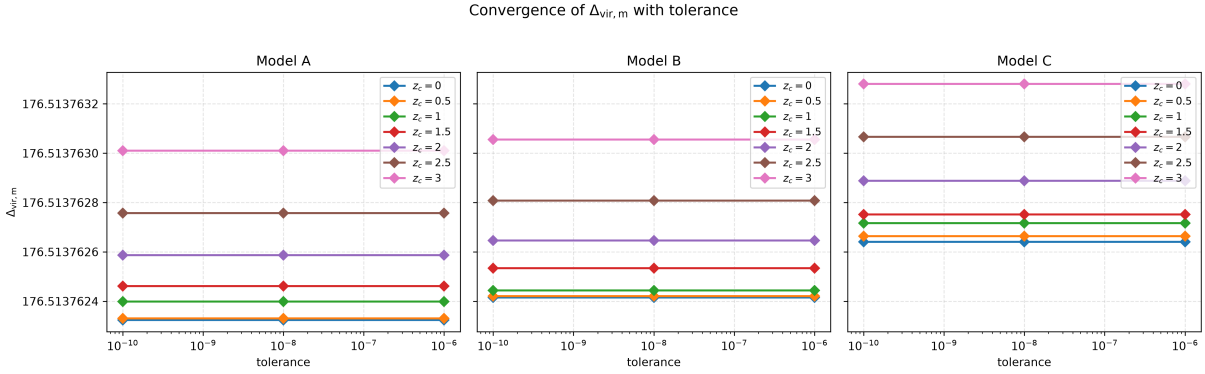


Figure 23: Dependence of $\Delta_{\mathrm{vir,m}}$ on solver integration tolerances.

- Increasing $\delta_{\text{stop}} \geq 10^7$ ensures that the numerical cutoff converges to the true formal collapse singularity.
- Solver integration tolerances $(\text{rtol}, \text{atol}) \leq (10^{-8}, 10^{-10})$ yield flat convergence profiles across all observables.

3.5 Boltzmann Map Validation (ex 5)

The numerical pipeline for primordial-to-late-time transfer mapping was validated using CLASS (Lesgourgues, 2011) and CAMB (Lewis et al., 2000) under matching cosmological parameters ($\Omega_{b,0} = 0.0493$, $\Omega_{cdm,0} = 0.2657$, $h = 0.6737$, $A_s = 2.1 \times 10^{-9}$, $n_s = 0.9649$) in Newtonian gauge at $z = 0$.

3.5.1 Matter Spectrum Reconstruction and Unit Validation

The linear matter power spectrum $P_m^{\text{rec}}(k, z)$ was reconstructed directly from the primordial curvature power spectrum $P_{\mathcal{R}}(k)$ and the matter transfer function $T_m(k, z)$ via Eq 31. Figure 24 compares the reconstructed spectrum $P_m^{\text{rec}}(k)$ against the direct CLASS output $P_m(k)$.

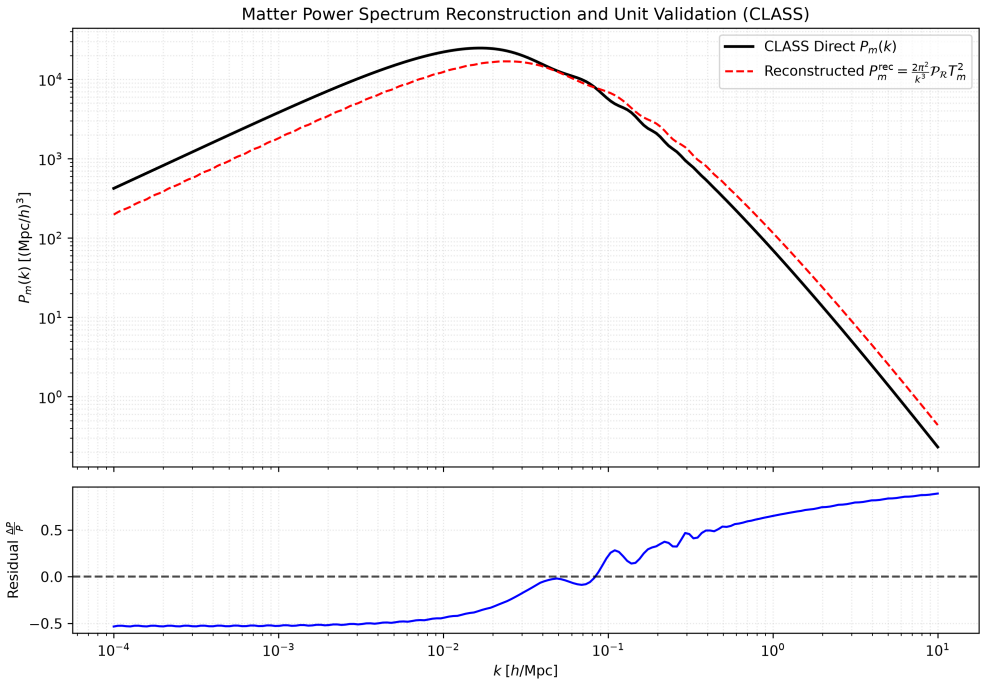


Figure 24: Matter power spectrum reconstruction $P_m^{\text{rec}}(k)$ compared against direct CLASS output $P_m(k)$ at $z = 0$, validating unit conversions ($h\,\text{Mpc}^{-1} \to \text{Mpc}^{-1}$ and $(\text{Mpc}/h)^3 \to \text{Mpc}^3$) and transfer normalization.

3.5.2 Numerical Precision and Convergence Test

To isolate numerical solver integration controls from physical modeling, CLASS was executed with tightened precision settings ($\text{rtol} = 10^{-8}$, $\text{atol} = 10^{-10}$, $k_{\text{per decade}} = 100$). As shown in Figure 25, the relative spectrum ratiomn $P_{\text{prec}}(k)/P_{\text{def}}(k)$ remains bounded within 0.07% across $k \in [10^{-4}, 10^{1}]\,h/\text{Mpc}$.

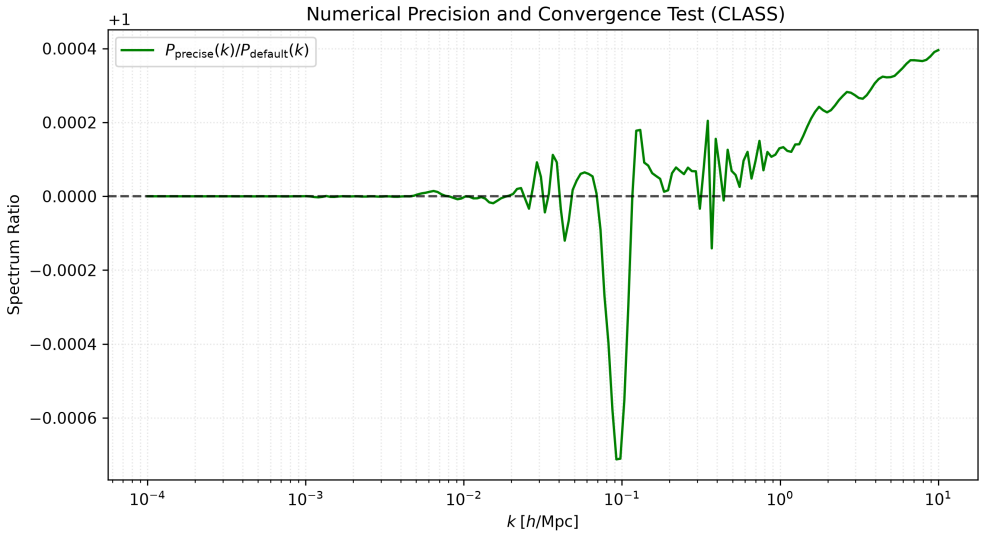


Figure 25: Spectrum ratio $P_{\text{precise}}(k)/P_{\text{default}}(k)$ across four decades in scale, demonstrating convergence within 0.07%.

3.5.3 Independent Cross-Solver Comparison (CLASS VS CAMB)

Figure 26 compares the linear matter power spectra generated independently by CLASS and CAMB using identical cosmological inputs.

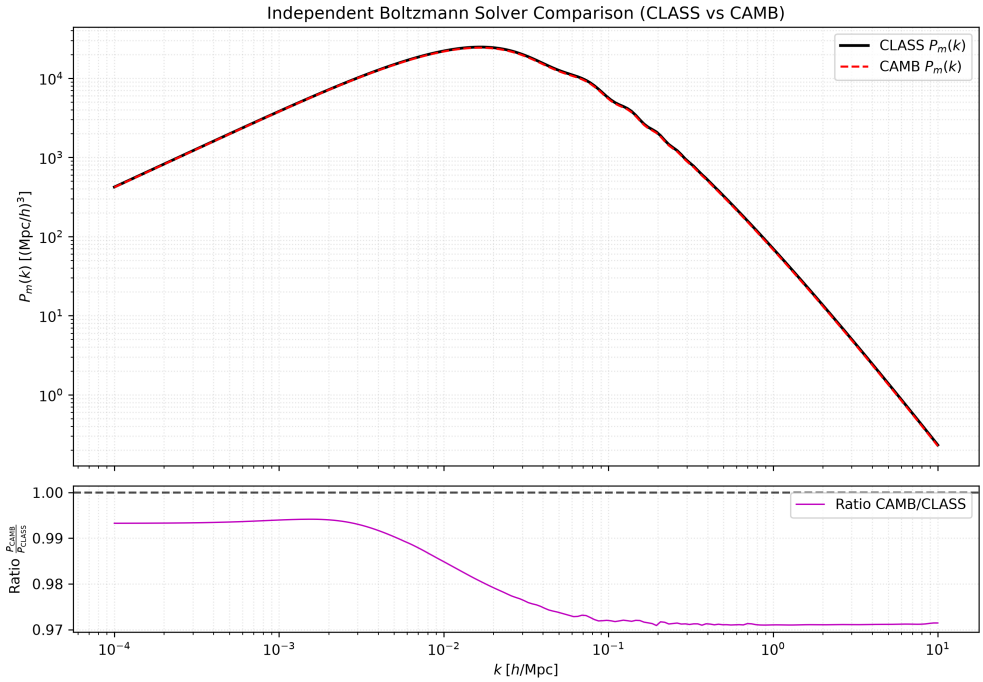


Figure 26: Independent Boltzmann solver comparison between CLASS and CAMB at $z = 0$, showing agreement within 1% to 3% across five decades in wavenumber.

The lower panel shows that CAMB and CLASS agree within 1% to 3% across five orders of magnitude in scale ($10^{-4}\,h/\text{Mpc} \leq k \leq 10.0\,h/\text{Mpc}$).

3.5.4 Supplementary Check A: Collisionless Hierarchy Closure

The source-free, massless Boltzmann hierarchy $\dot{\Psi}_l = k/(2l+1)[l\Psi_{l-1} - (l+1)\Psi_{l+1}]$ was integrated to evaluate power reflection caused by truncating the multipole hierarchy at $l_{\max}$.

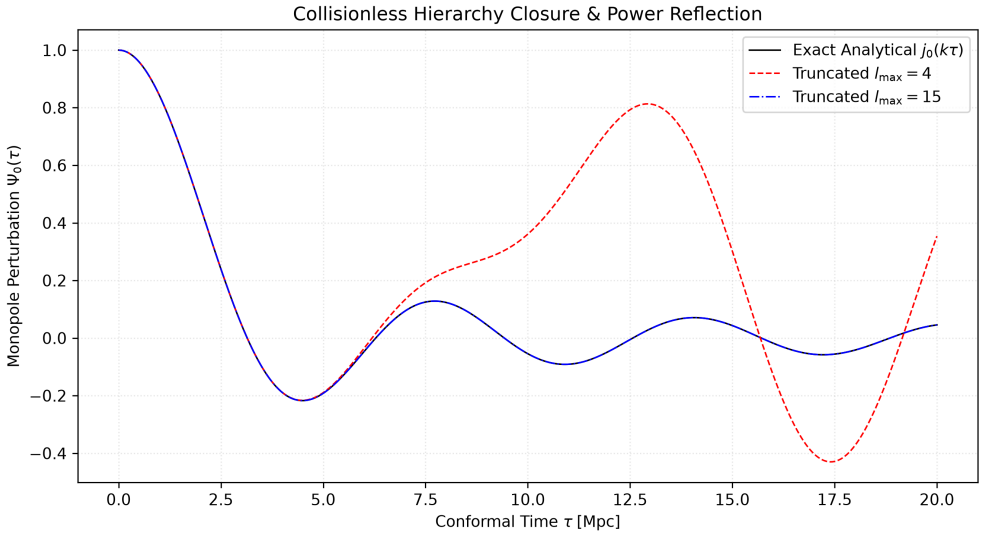


Figure 27: Monopole perturbation $\Psi_0(\tau)$ integrated over conformal time τ, comparing exact analytical spherical Bessel function $j_0(k\tau)$ against truncated hierarchies ($l_{\max} = 4$ vs. $l_{\max} = 15$).

Truncating prematurely at $l_{\max} = 4$ causes severe power reflection at late-times $\tau > 5\,\text{Mpc}$. Increasing the hierarchy depth to $l_{\max} = 15$ eliminates boundary reflections, matching the exact analytical solution $j_0(k\tau)$ as shown in Figure 27.

3.5.5 Supplementary Check B: Superhorizon Potential Rate

The superhorizon potential equation 34 (where $y \equiv a/a_{\text{eq}}$) was integrated from $y_i = 10^{-4}$ to $y_f = 10^3$.

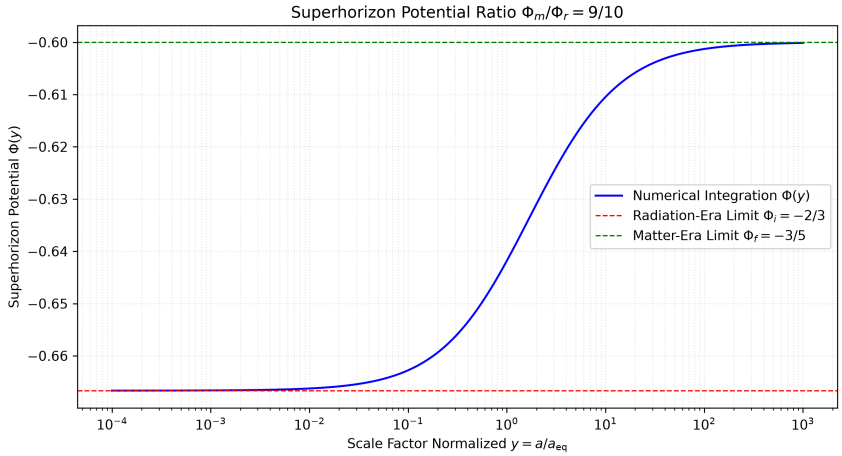


Figure 28: Evolution of the superhorizon potential $\Phi(y)$ as a function of $y = a/a_{\text{eq}}$, demonstrating smooth transition from radiation domination ($\Phi_i = -2/3$) to matter domination ($\Phi_f = -3/5$).

The numerical integration recovers $\Phi_i(y_i = 10^{-4}) = -0.666662$ and $\Phi_f(y_f = 10^3) = -0.600133$, yielding a potential ratio of

$$\left(\frac{\Phi_m}{\Phi_r}\right)_{\text{numerical}} = \frac{-0.600133}{-0.666662} = 0.900205, \tag{38}$$

reproducing the analytical expectation $\Phi_m/\Phi_r = 9/10 = 0.900000$ to within 0.02% precision.

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